Operational amplifiers

Higher · Electronics & control · Worksheet

ENGINEERING WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ identify inverting, non-inverting, summing, difference and comparator op-amp circuits
- ☐ calculate gain and output voltage, including saturation at about 85% of supply
- ☐ design an op-amp circuit with suitable resistor values
- ☐ describe how a comparator switches an output

I DO — Worked example

Read how a full-mark answer is built

The same amplifier (gain −10) now has Vin = 1.5 V. Supply ±12 V.

1 Calculate Vout, taking the supply into account. (2 marks)

Full-mark answer each ✓ is one mark

−10 × 1.5 = −15 V, which is beyond the supply ✓. The op-amp saturates at about 85% of 12 V, so Vout ≈ −10.2 V ✓.

Why marks are not awarded

"Vout = −15 V"

0 marks: ignores saturation.

WE DO — Guided practice

Use the starters and word bank to help you

An inverting amplifier has Rin = 4.7 kΩ and Rf = 47 kΩ. Vin = 0.35 V. Supply ±12 V.

2 Calculate Vout. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which configuration is used as a buffer?
☐ Summing ☐ Voltage follower ☐ Comparator ☐ Inverting

b) On a 12 V supply, an op-amp saturates at about…
☐ 12 V ☐ 10.2 V ☐ 6 V ☐ 0.7 V

c) Rf = 100 kΩ, Rin = 10 kΩ, Vin = 0.2 V (inverting). Vout is…
☐ 2 V ☐ −2 V ☐ 0.02 V ☐ −0.2 V

d) A comparator output goes high when…
☐ V− > V+ ☐ V+ > V− ☐ Both are equal ☐ Supply is off

4 Key terms: write the letter of the matching description next to each term.

1. **Inverting amplifier**	____	**A** Vout = (Rf/R1) × (V2 − V1)
2. **Non-inverting amplifier**	____	**B** Vout = (1 + Rf/R1) × Vin
3. **Summing amplifier**	____	**C** Output high when V+ > V−, otherwise low.
4. **Difference amplifier**	____	**D** Vout = −Rf × (V1/R1 + V2/R2 + …)
5. **Voltage follower**	____	**E** Vout = Vin; used as a buffer.
6. **Comparator**	____	**F** Vout = −(Rf/Rin) × Vin

A non-inverting amplifier must have a gain of 11. Rin = 1 kΩ.

5 Calculate Rf. (2 marks)

A summing amplifier has inputs of 0.2 V and 0.5 V, each through 10 kΩ, with Rf = 22 kΩ.

6 Calculate Vout. (2 marks)

A difference amplifier has V1 = 2.4 V, V2 = 3.0 V, R1 = 10 kΩ and Rf = 33 kΩ (balanced).

7 Calculate Vout. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?